

Lab: Feeling Tension in Your Fingers

Objective

Explore the sensory dimensions of crochet by systematically varying tactile conditions and observing how your perceptual system responds — detecting prediction errors, adjusting precision weighting, and discriminating between different sensory signals.

Materials

- Three different yarns: (1) smooth worsted acrylic, (2) textured or fuzzy yarn (boucle, mohair, or eyelash), (3) cotton or bamboo
- Crochet hook, size H/8 (5.0mm)
- Blindfold or sleep mask
- A partner or crochet circle companion (optional but recommended)
- Pen and Perception Log sheet (template below)
- Scissors

Perception Log Template

Trial	Condition	What I Expected	What I Felt	Surprise Level (1-5)	Notes

Steps

Part 1: Baseline — Eyes Open, Familiar Yarn (10 minutes)

1. Using the smooth worsted acrylic, chain 12 and work 3 rows of single crochet with your eyes open. This is your baseline.
2. In your Perception Log, note what you feel in your fingers: the tension, the resistance of each stitch, the weight of the hook, the texture of the yarn.

3. Rate how "automatic" this feels on a scale of 1-5 (1 = must think about every movement, 5 = completely automatic). This baseline captures your normal perceptual state.

Part 2: Eyes Closed — Proprioceptive Perception (10 minutes)

1. Put on the blindfold. Continue crocheting single crochet rows in the same swatch.
2. Work at least one full row blind. Notice: what sensory channels do you rely on now? What information are you getting from your fingers that you did not consciously notice with your eyes open?
3. Record in your log: What surprised you? Did you drop a stitch or lose your place? How did you detect the error without looking?
4. Remove the blindfold. Count your stitches. Record any discrepancy between expected and actual count.

Part 3: Yarn Texture Discrimination (15 minutes)

1. Set aside the acrylic swatch. Pick up the textured yarn and start a new swatch: chain 12, work 2 rows of single crochet.
2. Record how this yarn feels compared to the acrylic. What is different about the tension, the slide of the yarn, the resistance when pulling through?
3. Now switch to the cotton/bamboo yarn and repeat: chain 12, 2 rows of single crochet.
4. Record the sensory differences. Cotton tends to feel smoother and less stretchy; bamboo feels silky.
5. **Blind identification test:** Have a partner hand you one of the three yarns without telling you which one. With your eyes closed, handle it, try a few chains, and identify which yarn it is. Record whether you were correct and what sensory cues you used.

Part 4: Detecting Errors by Feel (15 minutes)

1. Return to your acrylic swatch, eyes open. Work a new row, but at some point **deliberately insert your hook under only one loop** instead of both loops of the stitch below.
2. Continue crocheting normally. Can you feel the difference? The single-loop insertion should produce a slightly different resistance. Record the sensation.

3. Now, work another row and **deliberately skip one stitch**. Continue to the end of the row. Did you feel the skip before you saw it? Record your observations.
4. Work one more row and **deliberately use inconsistent tension** — make one stitch very tight and the next very loose. Feel the irregularity. Record what your fingers tell you.

Part 5: Reflection and Discussion (10 minutes)

1. Review your Perception Log across all trials. In which condition was your surprise level highest? When was your perceptual system most strained?
2. Compare your eyes-open and eyes-closed performance. What did you learn about the role of vision versus touch in your crochet perception?

Discussion Questions

- Which sensory channel (vision, touch, proprioception, rhythm) carried the most information during eyes-closed crochet? Were you surprised by how much you could do without looking?
- When you switched yarns, did your stitching slow down? What does this tell you about the precision of your generative model for unfamiliar materials?
- Could you tell the difference between a single-loop insertion and a both-loops insertion by feel alone? What does this say about the resolution of your tactile generative model?
- In your crochet circle, compare perception logs with others. Do experienced crocheters report lower surprise levels? Do beginners rely more on vision?

Wrap-Up

You have explored the perceptual world of crochet: tension as proprioception, surprise as error detection, and precision weighting across different yarns and conditions. Keep your three mini-swatches and your Perception Log for reference in future modules.